BIOTECHNOLOGY

Floral waste valorization and sustainable resource management

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ABSTRACT:

Floral waste from religious offerings, markets, and ceremonies is an emerging environmental challenge. However, it also presents a valuable bioresource. This study explores the potential of fresh and waste flowers from *Tagetes erecta* (marigold) and *Rosa spp*. (rose) for extracting antioxidant and antimicrobial compounds. The flowers were collected from temples and marketplaces in Mumbai and categorized as fresh or waste. They were dried, powdered, and subjected to Soxhlet extraction using 95% ethanol to maximize the recovery of bioactive compounds such as flavonoids, phenolics, tannins, saponins, and terpenoids.

Qualitative tests confirmed the presence of these phytochemicals, though in varying amounts between fresh and waste flower extracts. Quantitative analysis through UV-visible spectroscopy revealed that fresh marigold had the highest flavonoid content (0.73 mg/mL), while waste rose had the highest phenolic content (0.71 mg/mL). Fresh marigold also recorded the highest tannin content (0.98 mg/mL).

The antioxidant activity was assessed using the DPPH radical scavenging assay, where waste marigold demonstrated the highest activity at 85.07% inhibition at a 100 µg/mL concentration, surpassing the standard ascorbic acid. Antimicrobial activities were tested against *Escherichia coli* and *Staphylococcus aureus* using the agar well diffusion method. Waste marigold extract showed the largest inhibition zone against *E. coli* (37 mm), while waste rose extract exhibited 31 mm

inhibition against the same bacterium. These results were comparable to the standard antibiotic ampicillin.

This study underscores the potential of floral waste as a source of bioactive compounds for pharmaceutical, cosmetic, and industrial applications. By valorizing floral waste, this research aligns with the principles of the circular economy and sustainable waste management. Future studies should optimize extraction techniques, evaluate the stability of bioactive compounds, and assess their efficacy through in vivo studies to explore commercial viability.

Keywords: valorization of floral waste, bioactive compounds, antioxidants, antimicrobial activity, Soxhlet extraction, sustainable resource management, circular economy

INTRODUCTION:

The increasing accumulation of floral waste from markets, temples, and cultural events poses serious environmental challenges, including microbial contamination, organic pollution, and ecosystem degradation. Commonly discarded in landfills or water bodies, floral waste contributes significantly to local pollution (Vasileva et al., 2019). However, flowers are rich in bioactive compounds such as flavonoids, phenolics, tannins, and terpenoids, which exhibit potent antioxidant, antimicrobial, and therapeutic activities (Agrawal et al., 2022). Valorizing floral waste presents a sustainable approach to convert biodegradable waste into value-added products, with applications in pharmaceuticals, food preservation, and cosmeceuticals.

This study focuses on comparing fresh and waste floral samples of *Tagetes erecta* (marigold) and *Rosa* spp. (rose) in terms of phytochemical composition, antioxidant capacity, and antimicrobial potential. Post-harvest exposure to microbial activity and oxidative stress may either degrade or enhance specific bio actives. Previous research indicates that enzymatic oxidation and microbial transformation can increase the bioavailability of certain compounds in waste flowers

(Ahmad et al., 2023), suggesting that waste flowers may retain or even improve in biochemical potency.

The study emphasizes flavonoids and phenolics due to their free radical-scavenging and antibacterial effects. Tannins and terpenoids, known for their antimicrobial properties, also contribute to the therapeutic value. Despite being discarded, these flowers may undergo biochemical transformations that enhance their bioactivity (Cahyaningrum & Widyantari, 2023).

Ethanol-based Soxhlet extraction was chosen for its efficiency in extracting a broad spectrum of phytochemicals while preserving bioactivity (Manimegalai & Selvam, 2023). Extracts were analyzed through qualitative and quantitative phytochemical screening, with UV-visible spectrophotometry used to estimate flavonoid and phenolic content. The antioxidant potential was assessed using the DPPH radical scavenging assay, which highlights whether waste flowers maintain or enhance their antioxidant activity. Some studies report that oxidative stress can trigger secondary metabolite production in plants, increasing antioxidant activity post-discard (Huang et al., 2022).

For antimicrobial testing, the agar well diffusion method was used against *Escherichia coli* and *Staphylococcus aureus*. Active compounds like flavonoids and tannins disrupt bacterial membranes and inhibit growth. Waste extracts may even show enhanced antibacterial action due to biotransformation.

Initial results suggest both fresh and waste flowers contain key bio - actives, though in varying concentrations. This study highlights the potential of floral waste as a viable source of natural antioxidants and antimicrobials, contributing to sustainable waste management and the circular bioeconomy in India.

MATERIALS AND METHODS:

A. Collection and Preparation of Samples

Fresh and waste flowers of *Tagetes erecta* (marigold) and *Rosa* spp. (rose) were collected from the Dadar flower market and a local temple in Mumbai. The flowers were sorted, cleaned, and dried using a hot air oven to preserve bioactive compounds. Dried samples were ground into a fine powder using a mechanical grinder and stored in airtight containers for further analysis.

B. Extraction of Bioactive Compounds

Ten grams of each powdered flower sample were subjected to Soxhlet extraction using 150 mL of 95% ethanol for 4 hours. Ethanol was selected for its efficiency in extracting diverse phytochemicals. The extract was then concentrated under reduced pressure and dried using a hot air oven to remove residual solvent.

C. Phytochemical Screening

1. Qualitative analysis included tests for alkaloids (Dragendorff's reagent), flavonoids (lead acetate), phenolics and tannins (ferric chloride), saponins (foam test), and terpenoids (Salkowski test).

2. Quantitative analysis was done using UV-Visible spectrophotometry: phenolics at 271 nm (gallic acid), tannins at 276 nm (tannic acid), flavonoids at 370 nm (quercetin), terpenoids at 530 nm, and saponins at 540 nm (Vallin-Sulphuric acid assay).

D. Antioxidant and Antimicrobial Activity

Antioxidant activity was tested using the DPPH assay. Extracts (20–100 µg/mL) were mixed with DPPH solution, and absorbance was measured at 517 nm. Ascorbic acid served as a standard and formula Percentage inhibition = (A control −A sample /A control)×100 was used. Antimicrobial activity was determined by the agar well diffusion method using *E. coli* and *S. aureus*. Wells filled with extract dilutions (pure, 10^{-1}, 10^{-2}) were incubated on Mueller-Hinton agar, and zones of inhibition were measured after 24–48 hours at 37°C. Ampicillin was taken as positive control and DMSO was taken as negative control.

RESULTS:

1. Selection, collection and Processing of flowers

Fig 1.1 Fresh Marigold and Rose collected from Temple and Dadar market in Mumbai

Fig 1.2 Fresh Marigold and Rose collected from Temple and Dadar market in Mumbai

2. Extraction Process

Fig 1.3 Soxhlet Extraction

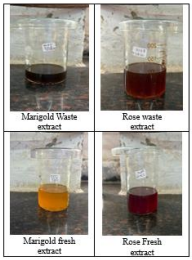


Fig 1.4 Extracted Samples

3. Qualitative analysis of phytochemicals

Qualitative analysis was performed to detect the presence of phytochemicals such as flavonoids, phenolics, tannins, saponins, terpenoids, and alkaloids in the floral extracts. Standard chemical tests based on colour change and foam formation were used to confirm their presence.

Components	Test	Expected Results	Fresh Marigold	Fresh Rose	Waste Marigold	Waste Rose
Alkaloids	Dragendroff's Reagent	Orange Precipitate	+	-	+	+
Flavonoids	Lead Acetate Test	Yellow precipitate	+	-	+	+
Tannins	Ferric Chloride Test	Blue, greenish or black colour	+	+	+	+
Phenolics	Ferric Chloride Test	Blue, greenish, or black colour	+	+	+	+
Terpenoids	Salkowski Test	Reddish-brown colour at interface	+	+	+	+
Saponins	Foam Test	Persistent foam formation	+	+	+	-

Key:	
+	Positive results
-	Negative results

Table 1: Protocol and Observation table for qualitative analysis of presence of various bioactive compounds

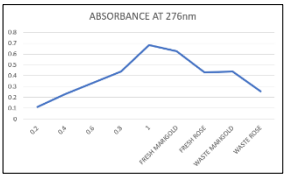
4. Quantitative Analysis

Quantitative analysis was conducted using UV-Visible spectroscopy to estimate the concentration of bioactive compounds such as flavonoids, phenolics, tannins, saponins, and terpenoids. Standard calibration curves were used to determine the precise amounts of each compound in the flower extracts.

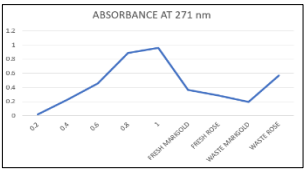
Flower Extract	Concentration (mg/ml)				
Phytochemicals:	Tannins	Phenolics	Flavonoids	Terpenoids	Saponins
Fresh marigold	0.98mg/ml	0.25mg/ml	0.73mg/ml	0.47mg/ml	0.31mg/ml
Fresh rose	0.76mg/ml	0.42mg/ml	0.69mg/ml	0.15mg/ml	0.31mg/ml

Waste marigold	0.78mg/ml	0.39mg/ml	0.56mg/ml	0.13mg/ml	0.22mg/ml
Waste rose	0.47mg/ml	0.71mg/ml	0.42mg/ml	0.15mg/ml	0.36mg/ml

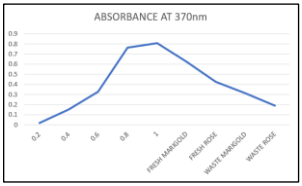
Table 2: Concentrations of phytochemicals in Flower extracts



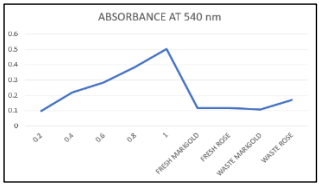
Graph 1: Concentration of Tannins



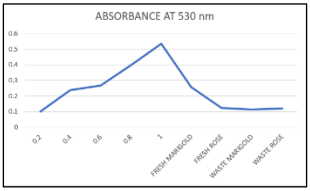
Graph 2: Concentration of Phenolics



Graph 3: Concentration of Flavonoids



Graph 4: Concentration of Saponins



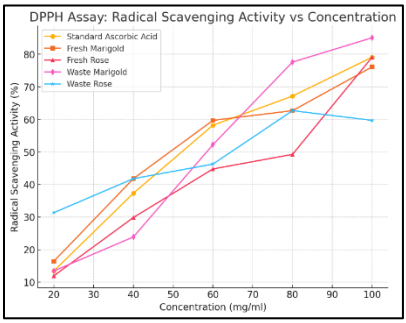
Graph 5: Concentration of Terpenoids

5. Antioxidant Analysis

Antioxidant activity of the flower extracts was evaluated using the DPPH radical scavenging assay. The percentage of free radical inhibition was measured spectrophotometrically to assess the extracts' ability to neutralize oxidative stress.

Concentration (mg/ml)	Standard Ascorbic Acid (%)	Fresh marigold (%)	Fresh rose (%)	Waste marigold (%)	Waste rose (%)
20	13.43	16.24	11.94	13.43	31.34
40	37.31	41.79	29.85	23.88	41.79
60	58.21	59.70	44.78	52.24	46.27
80	67.16	62.69	49.25	77.61	62.69
100	79.10	76.12	79.10	85.07	59.70

Table 3: Antioxidant activity (Radical Scavenging Activity [%]) of the flowers extract after DPPH assay



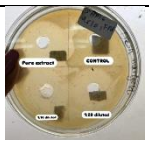
Graph 6: DPPH Radical Scavenging Activity in flower extracts

6. Antimicrobial Assay

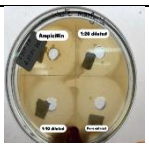
Antimicrobial assay was performed using the agar well diffusion method to evaluate the antibacterial activity of the flower extracts. The extracts were tested against *Escherichia coli* ATCC25922 and *Staphylococcus aureus* ATCC29213, and zones of inhibition were measured to determine their antimicrobial effectiveness.

	Zone of inhibition			
Dilutions	Fresh marigold	Fresh Rose	Waste Marigold	Waste Rose
Pure Extract	34 mm	27mm	37mm	31mm
1:10 diluted	26 mm	23mm	25mm	27mm
1:20 diluted	23 mm	22mm	21mm	26mm
Ampicillin	-	-	25mm	25mm
DMSO	0mm	0mm	-	-

Table 4: Measurement of zone of inhibition of antimicrobial Activity of flower extracts against *Escherichia coli* ATCC25922

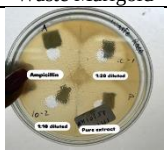


Fresh Marigold



Waste Marigold

Fresh Rose



Waste Rose

Fig 1.5 Antimicrobial Activity of flower extracts against *Escherichia coli* ATCC25922

	Zone of inhibition			
Dilutions	Fresh marigold	Fresh Rose	Waste Marigold	Waste Rose
Pure Extract	27mm	20mm	25mm	21mm
1:10 diluted	16mm	16mm	13mm	17mm
1:20 diluted	15 mm	14mm	12mm	12mm

Ampicillin	-	-	27mm	26mm
DMSO	0mm	0mm	-	-

Table 5: Measurement of zone of inhibition of antimicrobial Activity of Fresh marigold flower extract against *Staphylococcus aureus* ATCC 29213

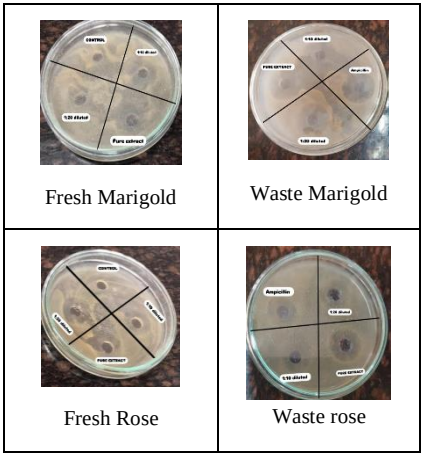


Fig 1.6 Antimicrobial Activity of flower extracts against *Staphylococcus aureus* ATCC 29213

DISCUSSION:

This study evaluated the phytochemical composition, antioxidant capacity, and antimicrobial activity of fresh and waste flower extracts from *Tagetes erecta* (marigold) and *Rosa* spp. (rose), with particular focus on the differences in bioactivity due to extraction methods and sample type. The use of Soxhlet extraction with 95% ethanol proved efficient for recovering polar and semi-polar bio actives such as flavonoids, phenolics, tannins, and terpenoids. Waste flowers, especially those sourced from worship sites, exhibited higher phenolic content and

antioxidant activity such as waste marigold's 85.07% inhibition in DPPH assays likely due to microbial transformation and post-harvest metabolic changes. This agrees with reports suggesting oxidative stress and microbial fermentation can enhance secondary metabolite accumulation.

Quantitative analyses showed significant variations compared to qualitative screenings. While all extracts tested positive for bio actives qualitatively, the actual concentrations varied, reinforcing the need for UV-visible spectrophotometry in phytochemical analysis. For instance, fresh marigold had a higher flavonoid content than waste marigold, possibly due to degradation during environmental exposure.

Antimicrobial testing revealed that waste marigold extract had the highest inhibition zone (37 mm) against *E. coli*, surpassing the standard antibiotic ampicillin. This indicates that waste flower extracts may yield potent antimicrobial agents through biotransformation. However, over-incubation (48h) could have artificially inflated inhibition zones, stressing the importance of standardized protocols.

The presence of alkaloids in waste rose and their absence in fresh rose suggests microbial or enzymatic conversion of compounds during degradation. Overall, the study supports the valorization of floral waste in sustainable biotechnology, highlighting its potential in developing antioxidant supplements, natural preservatives, and plant-based antimicrobials contributing to eco-friendly solutions and circular economy.

CONCLUSION:

This study highlights the potential of floral waste, specifically marigold and rose, as valuable sources of bioactive compounds with significant antioxidant and antimicrobial properties. Soxhlet extraction using ethanol proved efficient in isolating flavonoids, phenolics, tannins, and terpenoids. Waste marigold showed strong antioxidant activity (85.07% DPPH inhibition) and potent antibacterial effects, notably against *E. coli* and *S. aureus*. Waste rose also demonstrated comparable phytochemical

content and antimicrobial potential. These findings align with previous research, reinforcing the suitability of ethanol as a solvent and the therapeutic potential of these extracts.

Valorizing floral waste supports circular economy principles by converting environmental waste into functional ingredients for pharmaceuticals, cosmetics, and food preservation. The study suggests that even discarded flowers retain significant bioactivity, making them sustainable alternatives to synthetic compounds. Future work should explore advanced extraction methods, optimize formulations, and conduct in vivo studies to assess commercial viability. Overall, this research supports the development of natural, waste-derived bioactive products, contributing to both environmental sustainability and human health.

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